

Ardeshir Aidun

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PROFESSIONAL SUMMARY

Electrical Engineering student at UC Santa Barbara with hands-on experience in embedded systems, FPGA design, circuit prototyping, and bench testing. Proficient in SystemVerilog, C++, and Python, with a track record of designing and validating hardware systems from concept through testing and deployment.

EDUCATION

University of California, Santa Barbara

Bachelor of Science, Electrical Engineering. GPA: 3.72.

Expected June 2028

Relevant Coursework: Digital Design, Analog and Digital Circuits Fundamentals, Signal Analysis, Robotics: Control, Quantum Photonics, Research Process for Engineers

PROJECTS

ESP32 IoT Mesh Network: Wireless Telemetry and Rollback-Safe OTA Updates

June 2026 – July 2026

- Designed and built a 3-node wireless mesh network using ESP32 microcontrollers, with sensor nodes reporting live temperature and humidity data to a central gateway over ESP-NOW
- Architected a wireless OTA firmware update system using WiFi and HTTP, leveraging the ESP32's dual-partition bootloader to safely stage and verify updates without disrupting the running device
- Validated fault-tolerant recovery by deploying broken firmware and confirming automatic rollback to the last known-good version with zero manual intervention

1965 Thunderbird Taillight Controller (FPGA)

Mar. 2026 – Mar. 2026

- Designed an 8-state Moore finite state machine in SystemVerilog to replicate the sequential taillight behavior of a 1965 Ford Thunderbird, covering turn signal, brake, and hazard functions
- Implemented priority-encoded control logic to resolve input conflicts among simultaneous turn signal, brake, and hazard inputs per a strict override specification
- Synthesized and deployed the design to an FPGA, verifying correct LED output across all input combinations against a truth table

Ultrasonic Sensor Accuracy and Noise Characterization

May 2026 – June 2026

- Designed and executed a bench test protocol to characterize accuracy, noise, and drift of an HC-SR04 sensor across a 5-30 cm range using an Arduino Uno
- Built a Python data pipeline to log serial sensor data and compute statistical metrics (mean error, standard deviation, drift rate), visualized with matplotlib
- Identified a distance-dependent measurement bias and noise discrepancy between test protocols, documenting findings in a written report

Bike Indicator Lights

Oct. 2024 – Dec. 2024

- Designed and built a custom microcontroller system using an Arduino Uno and push buttons to control left, right, and hazard light functions on an LED strip
- Programmed custom control logic for directional signaling and emergency flashers

EXPERIENCE

Member

Oct. 2024 – Present

IEEE - UCSB Chapter

Goleta, CA

- Practiced soldering techniques and PCB design concepts through hands-on technical workshops
- Built and troubleshooted circuits, developing practical skills in component assembly and prototyping

Guest Advocate

June 2025 – June 2026

Target

San Jose, CA

- Coordinated order pickups and curbside deliveries, resolving fulfillment and self-checkout issues

Leader

Aug. 2022 – May 2024

Link Crew

Los Gatos, CA

- Mentored incoming students through orientation and campus acclimation activities

SKILLS

Hardware: Microcontrollers, FPGA, Circuit Design, PCB Design, SystemVerilog, Bench Testing, Oscilloscope, Soldering, Prototyping, LTspice

Communication Protocols: UART, ESP-NOW, WiFi/HTTP

Software and Programming: C++, Python, MATLAB, Simulink, Git/GitHub, Technical Documentation

Technical Concepts: Digital Design, Signal Analysis, Finite State Machines, Boolean Algebra